
Platyhelminthes are triploblastic, bilaterally symmetrical, dorso ventrally flattened, acoelomate flatworms with organ grade of construction without a definite anus, circulatory, skeletal or respiratory system but with Protonephridia excretory system and mesenchyme filling the space between the various organ of the body.

Classification

- ✦ Class Turbellaria (Free-Living Flatworms)
- ✦ Class Monogenea
- ✦ Class Trematoda (Flukes)
- ✦ Class Cestoidea (Tapeworms)

88.Characteristics of Mammalian?

- ✦ Mammary gland
- ✦ Specialized teeth
- ✦ Skin is thick and protective
- ✦ Three middle ear Ossicles
- ✦ Parental care
- ✦ Have hair
- ✦ Endothermic

114-Types of ecological pyramids?(10ppt)

Ecological Pyramid refers to a graphical (pyramidal) representation to show the number of organisms, biomass, and productivity at each trophic level.

There are three different types of ecological pyramids to illustrate ecosystems:

1. **Pyramid of energy**: represents how much energy is available in each trophic level
2. **Pyramid of numbers**: represents the actual number of organisms present in each trophic level
3. **Pyramid of biomass**: represents the total mass of living things in each trophic level

94-Different types of pseudopodia?(130ppt)

Types of pseudopodia

Majorly amoeba has four types

1. Lobopodia
2. Filopodia
3. Reticulopodia
4. Axopodia

95-What is cytoskeleton? Amoeba (105ppt)

2. Relationship between evolution and paleontology 5 MARKS

Paleontology is the study of fossils. It provides some of the most direct evidence for evolution.

Fossils (to dig) are evidence of plants and animals that existed in the past and have become incorporated into the earth's crust.

Nicolas Steno (1638-86) Father of geology and stratigraphy James Hutton

Evolution of earth must be old .Observable processes produce small changes that accumulate over time

3. Types of grazer 2

Types of Grazing

1. Graminivory

Feeding primarily on grass. Specifically "true" grasses in family Poaceae

Examples: Horses, cattle, hippopotamus, grasshoppers

2. Coprophagy

Feeding on grass, forbs, leafy weeds, fruits, tree bark

Heavy grazing for half an hour of grazing period

Followed by half an hour of selective feeding

- Later feeding in intervals
- Eat their or other species's soft feces (caecotrophs)

Example: Rabbit

4. Difference between grazing and browsing ? 5

1. Grazers

Feeds on plants such as grasses, soft flowering plants (forbs) and algae. Plants are not killed.

Examples: Deer & Snails

Types of Grazing

1. Graminivory

Feeding primarily on grass. Specifically "true" grasses in family Poaceae

Examples: Horses, cattle, hippopotamus, grasshoppers

Domain Archaea

1. Methanogens

- They release methane (CH₄) as a waste product
- Many live in mud at the bottom of lakes and swamps because it lacks oxygen
- Some live in the intestinal tracts of animals to help break down food.
- They could play a role in garbage/sewage cleanup by eating garbage.

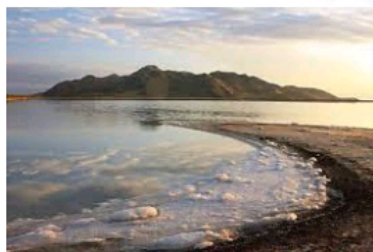
Domain Archaea

2. Thermoacidophiles

- Live in the dark
- Live without oxygen
- Like to live in superheated water with temperatures reaching 750°F
- Prefer environments that are very acidic (between pH of 1-3)
- Live in a chemical soup of hydrogen sulfide (H₂S) and other dissolved minerals.

Domain Archaea

3. Halophiles



<https://www.google.com/search?rlz=1C1SQJL>

- Halo = salt
phil = loving
- Can live in water with salt concentrations exceeding 15%.
- The ocean's concentration is roughly 4%.
- One habitat of halophils is Great Salt Lake.
- actually three to five times saltier than the ocean.



Domain Archaea

Logistic Population Growth

EXPONENTIAL GROWTH	LOGISTIC GROWTH
It occurs when resources are abundant.	It occurs when resources are limited.
Population passes well beyond the carrying capacity of the ecosystem.	Population doesn't grow beyond the carrying capacity of the ecosystem.
It produce J shaped curve.	It produce S shaped curve.
A stationary or stable phase is seldom achieved.	A stationary or stable phase is achieved.
It has two phases lag and log.	It has four phases lag, log, deceleration & steady .
It occurs in fewer organisms like lemmings, algal blooms, etc.	It is more common, example members of wildlife, yeast (a microscopic fungus), etc.

<https://www.youtube.com/watch?v=LYoguWiHaow>

Subphylum Sarcodinia

Types of pseudopodia

- Majorly amoeba has four types
1. Lobopodia
 2. Filopodia
 3. Reticulopodia
 4. Axopodia



a.



b.

The **University** of Wisconsin–Madison Arboretum has pioneered restoration ecology.

- a. The restoration of the prairie was at an early stage in November 1935.
- b. The prairie as it looks today. This picture was taken at approximately the same location as the 1935 photograph

Adaptations

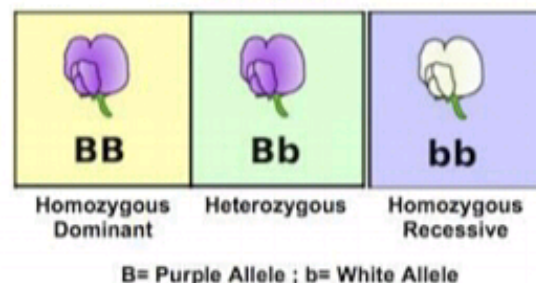
- Adaptation is confused with **fitness** or adaptedness.
- **Adaptedness or fitness:**
is a measure of the capacity for successful reproduction in a given environment.
- Not every characteristic is an adaptation to some kind of environmental situation.
- Neither adaptations lead to perfection.

END

Population Size, Genetic Drift & Neutral Selection

- Genes of all traits are present in a population in form of alleles.

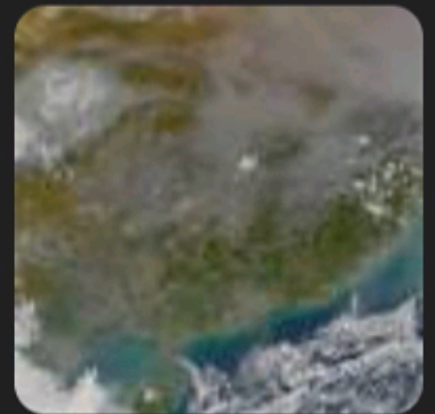
Example:



What is a pollution cloud?



atmospheric brown cloud, a layer of air pollution containing aerosols such as soot or dust that absorb as well as scatter incoming solar radiation,



leading to regional and global climatic effects and posing risks to human health and food security. 07-Aug-2023

31. Explain the characteristics of **nematodes**

- They are widely distributed, aquatic or terrestrial, parasitic or free-living.
- Their body is elongated, cylindrical, unsegmented, worm-like, bilaterally symmetrical and tapering at both ends.
- They are triploblastic animals with perivisceral cavity more extensive than that of **platyhelminths**.
- The body is of organ -system grade organization.
- The body is generally covered with thick, flexible multi-layered collagenous cuticle and often bears cuticle setae (hairs), spines or annulations.
- Cuticle moulted periodically.
- They have cellular or syncytial epidermis i.e. the nuclei are not separated from each other by cell membranes.
- They consist of only longitudinal muscle fibers with four bands.
- They lack true coelom. The body cavity is pseudocoel or blastocoel not lined by mesoderm and filled with parenchyma in most cases.
- They lack cilia.

Consumers

Heterotrophs

- Cannot synthesize organic compounds from inorganic precursors.
- Eat organisms from a different population for energy
- Liberating chemical-bond energy for metabolic use in glycolysis, Krebs's cycle and electron transport chain

Example: Animals

Consumers

Types of Consumers

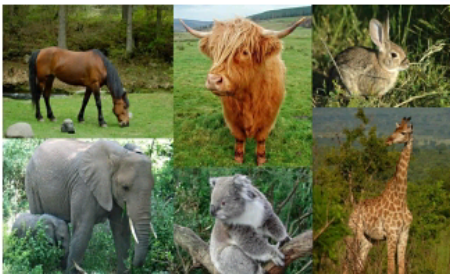


<https://www.popsci.com/most-powerful-animals-ranked>

1. Herbivore
2. Carnivore
3. Omnivore

Consumers

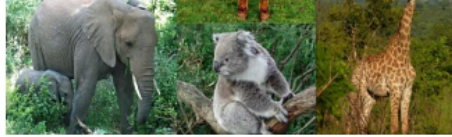
1. Herbivore



<https://thevetisin.com/carnivores-omnivores-herbivores/>

- Primary food source is plant-based.
- Examples: vertebrates as well as invertebrates
- Digestive systems capable of handling large amounts of plant material.

- There are eight types of herbivory depending upon different feeding habits.



<https://thetvetisin.com/carnivores-omnivores-herbivores/>

- Examples: vertebrates as well as invertebrates
- Digestive systems capable of handling large amounts of plant material.
- There are eight types of herbivory depending upon different feeding habits.

11

Consumers

Feeding Strategy	Diet	Example
Algivores	Algae	krill, crabs, sea snail, sea urchin, parrotfish, surgeonfish, flamingo
Frugivores	Fruit	Ruffed lemurs
Folivores	Leaves	Koalas
Nectarivores	Nectar	Honey possum
Granivores	Seeds	Hawaiian honeycreepers
Palynivores	Pollen	Bees
Mucivores	Plant fluids	Aphids
Xylophages	Wood	Termites

Consumers

2. Carnivore



<https://sco.wikipedia.org/wiki/Carnivora>

- Primary food source is animal-based.
- Examples: vertebrates as well as invertebrates
- Digestive systems is more efficient in protein digestion.

12

Consumers

2. Omnivore

- Primary food source is

179. Metamorphosis and its types. (5)

Metamorphosis, in biology, striking change of form or structure in an individual after hatching or birth. Hormones called molting and juvenile hormones, which are not species specific, apparently regulate the changes.

Metamorphosis is a process by which animals undergo extreme, rapid physical changes some time after birth. The result of metamorphosis may be change to the [organism](#)'s entire body plan, such as a change in the animal's number of legs, its means of eating, or its means of breathing. In [species](#) that use metamorphosis, metamorphosis is also typically required for sexual maturity. Pre-metamorphic members of these species are typically unable to mate or reproduce.

Types of Metamorphosis

Complete Metamorphosis

In [complete metamorphosis](#), a larva completely changes its body plan to become an adult. The most famous example is that of the butterfly, which starts out as a worm-like, [leaf](#)-eating caterpillar and transforms into a flying, nectar-drinking creature with an [exoskeleton](#).

Organisms that undergo complete metamorphosis are called "holometabolous," from the Greek words "holo" for "complete" or "whole," "meta" for "change," and the noun "bole" for "to throw." "Holometabolous," then, means "completely changing," or "wholly changing."

This transformation is so swift and complete that the caterpillar must spin a cocoon and lie dormant for weeks while its body undergoes these radical changes.

Other animals which transform from a worm-like larval stage into an animal that looks completely different include beetles, flies, moths, ants, and bees.

Incomplete Metamorphosis

In incomplete metamorphosis, only some parts of the animal's body change during metamorphosis. Animals that only partially change their bodies as they mature are called "hemimetabolous," from the Greek words "hemi" for "half," "meta," for "change," and the verb "bole" for "to throw."

"Hemimetabolous," then, is a word meaning "half-changing."

Cockroaches, grasshoppers, and dragonflies, for example, hatch from eggs looking a lot like their adult selves. They do acquire wings and functioning reproductive organs as they grow, but they do not completely remake their bodies like their completely metamorphosing cousins do.

180. Gene flow in a ecosystem. (5)

Exponential Population Growth

Exponential Growth

- Resources are abundant.
- Remarkable potential to increase in number
- Exponential addition in population size

Exponential Population Growth

The number of births minus the number of deaths per generation time

- Population may grow exponentially to certain time
- Ultimately limited by resource availability
- Formula of exponential growth

$$\frac{dN}{dt} = rN \quad N = \text{Population size}$$

r = intrinsic rate of natural increase

a suitable arrangement when it has the correct amount of all or three.

39. component of water pollutants

It includes detergents and other cleaning materials, surfactants, pesticides and other organic industrial wastes. Many of these materials are poisonous for living organisms and cause serious **water pollution** problems.

40. Overview of vestigial organs

END

- water at any given time.
- Responsible for water cycle
- Transfer water vapors from oceans to dry areas.
- Responsible for precipitation.
- Winds control weather.

Carbon Cycle

- Carbon is main component of living organisms.
- 20% of human body is carbon by weight.
- CO₂ is major source of carbon.
- Atmosphere contains about 750 billion metric tons of carbon in form of CO₂.

Carbon Cycle

- Photosynthetic organisms take in CO₂ from the atmosphere
- Synthesize the carbon containing organic compounds
- This process is sometimes called carbon fixation.

- Animals eat plants and make their own organic

containing organic compounds

- This process is sometimes called carbon fixation.

Carbon Cycle

- Animals eat plants and make their own organic compounds.
- Production of energy by breakdown of these organic compounds in respiration.
- Respiration releases CO_2 in air.
- Decaying organisms also produce CO_2 .

Carbon Cycle

- CO_2 released in air is again available for plants.
- In aquatic ecosystem, CO_2 reacts with the water to form bicarbonate ions.
- Dissolved CO_2 and bicarbonates are used by algae and aquatic plants in photosynthesis.

Carbon Cycle



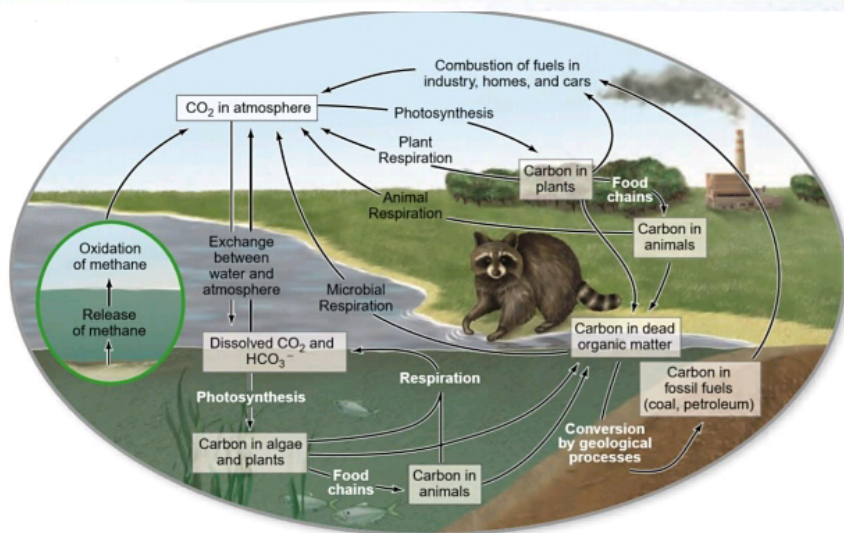
Methanogens, for example, are microbes that produce methane

- In aquatic ecosystem, CO_2 reacts with the water to form bicarbonate ions.
- Dissolved CO_2 and bicarbonates are used by algae and aquatic plants in photosynthesis.

Carbon Cycle

- Methanogens, for example, are microbes that produce methane (CH_4) instead of CO_2 .
- Methane released in air is oxidized into CO_2 .
- Amount of CO_2 made by breakdown of organic compounds almost matches the amount of CO_2 used.
- Small mis-matches can have large consequences.

Carbon Cycle



Biology Mason, Losos and Singer, 9th ed.

36. Types of omnivores

An **omnivore** is a kind of animal that eats either other animals or plants. Some **omnivores** will hunt and eat their food, like carnivores, eating herbivores and other **omnivores**. Some others are scavengers and will eat dead matter. Many will eat eggs from other animals.

Omnivores generally occupy the third trophic level alongside **meat**-eating carnivores. Omnivores are a diverse group of animals. Examples of omnivores include bears, birds, dogs, raccoons, foxes, certain insects, and even humans.

Molecular Evolution

- **Cytochrome c** is a protein present in the cellular respiration pathways in all eukaryotic organisms.
- Closely related organisms have similar *cytochrome c* molecules.
- Structure of *cytochrome c* has changed so little during hundreds of millions of years.
- It is said to have been conserved evolutionarily.

Molecular Evolution

- Highly conserved proteins can help establish evolutionary relationships among distantly related organisms.
- Not all proteins are conserved as rigorously as *cytochrome c*.
- Less conserved proteins are useful for looking at relationships among more closely related animals.

END

Regeneration in echinoderms:

All **echinoderms** have the capability to regenerate some or most organs or appendages. Since the regenerated structures are usually identical to the lost ones, including limbs, disc, gut, spines and podia and, in some species, regeneration is used for asexual reproduction. It has then been assumed that formation of new muscle cells or myogenesis occurs in all echinoderms.

r-selected species

- High growth rates
- Typically exploit less crowded ecological habitats
- Produce many offspring
- Relatively low probability of surviving to adulthood
- High reproductive rate supports their survival.
- Especially in unstable environment

Traits of r-selected species

- High rate of reproduction
- Small body size
- Early maturity onset
- Short generation time
- Ability to disperse offspring widely

Examples: bacteria, grasses, cephalopods, rodents

k-selected species

- Living at densities close to carrying capacity
- Typically are strong competitors in such crowded habitats.

MUHAMMAD

- Invest more heavily in fewer offspring
- Relatively high probability of surviving to adulthood
- Survive in stable environment
- Ability to compete successfully for limited resources is crucial
- Populations of k-selected organisms typically are very constant in number.
- Close to the maximum that the environment can bear

Traits of k-selected species

- Large body size
- Long life expectancy
- Production of fewer offspring
- Often require extensive parental care until they mature

Examples: humans, whales, parrots, eagles



<https://www.thinglink.com/scene/630237835143675906>



<https://www.worldatlas.com/articles/what-is-aestivation-animals-that-aestivate.html>

Aestivation

- Survival in extended period of drying
- Avoid temperature damage and dehydration
- Examples:
Insects, frog, turtle

Stages of **Water cycle**

1. Evaporation
2. Transpiration
3. Condensation
4. Precipitation
5. Surface Run off
6. Infiltration

Evaporation

In evaporation, water changes from its liquid state to a gaseous state. Some of the water in the oceans and freshwater bodies, such as lakes and rivers, is warmed by the sun and evaporates. During the process of evaporation, impurities in the water are left behind. As a result, the water that goes into the atmosphere is cleaner than it was on Earth.

Condensation

- Opposite of evaporation
- Condensation occurs when a gas is changed into a liquid. When the water droplets formed from condensation are very small.
- They remain suspended in the atmosphere. These millions of droplets of suspended water form clouds in the sky or fog at ground level.

Precipitation

- Under right temperature and atmospheric pressure conditions
- Small droplets of water in clouds form larger droplets and precipitation occurs.
- The raindrops fall to Earth.

Surface Runoff

Much of the water that returns to Earth as precipitation runs off the surface of the land

Flows downhill into streams, rivers, ponds and lakes. Small streams flow into larger streams, then into rivers, and eventually the water flows into the ocean. Surface runoff is an important part of the water cycle. Much of the water returns again to the oceans, where a great deal of evaporation occurs.

Layers of Atmosphere

- Atmosphere can be divided into five layers.
- Highest to lowest

Layers of atmosphere

1. Exosphere: 700 to 10,000 km (440 to 6,200 miles)
2. Thermosphere: 80 to 700 km (50 to 440 miles)[12]
3. Mesosphere: 50 to 80 km (31 to 50 miles)

Layers of Atmosphere

4. Stratosphere: 12 to 50 km (7 to 31 miles)
5. Troposphere: 0 to 12 km (0 to 7 miles)

Effects of Global Warming

Following are the major effects of global warming:

Rise in Temperature

Global warming has led to an incredible increase in earth's temperature. Since 1880, the earth's temperature has increased by ~1 degrees. This has resulted in an increase in the melting of glaciers, which have led to an increase in the sea level. This could have devastating effects on coastal regions.

Threat to the Ecosystem

Global warming has affected the coral reefs that can lead to a loss of plant and animal lives. Increase in global temperatures has made the fragility of coral reefs even worse.

Climate Change

Global warming has led to a change in climatic conditions. There are droughts at some places and floods at some. This climatic imbalance is the result of global warming.

Spread of Diseases

Global warming leads to a change in the patterns of heat and humidity. This has led to the movement of mosquitoes that carry and spread diseases.

Decrease in the Human Population

Due to an increase in floods, tsunamis and other natural calamities, the population of humans tend to decrease. Also, the spread of diseases leads to a decrease in the human population.

Loss of Natural Habitat

A global shift in the climate leads to the loss of habitats of several plants and animals. In this case, the animals need to migrate from their natural habitat and many of them even get extinct. This is yet another major impact of global warming on [biodiversity](#).

Evidence of the natural selection?

Darwin based his theory on many solid evidences from his observations and collected samples.

Three major evidences:

- ❖ Geology
- ❖ Fossil Evidences
- ❖ Galápagos islands

Land Pollution - 1

Pollution

Classified as:

1. Point source
2. Non point source

1. Point source

Single identifiable source of pollution

Example:

- Land fill
- Factory drain
- Sewages pipes

Land Pollution - 1

Pollution

Non Point source

Resulting from many diffuse sources

Example:

- Sediment
- Nutrients
- Toxic contaminants and chemicals
- Pathogens

Land Pollution - 1

Types of Pollution

1. Land
2. Air
3. Water
4. Light
5. Noise



Domain Eukaryota

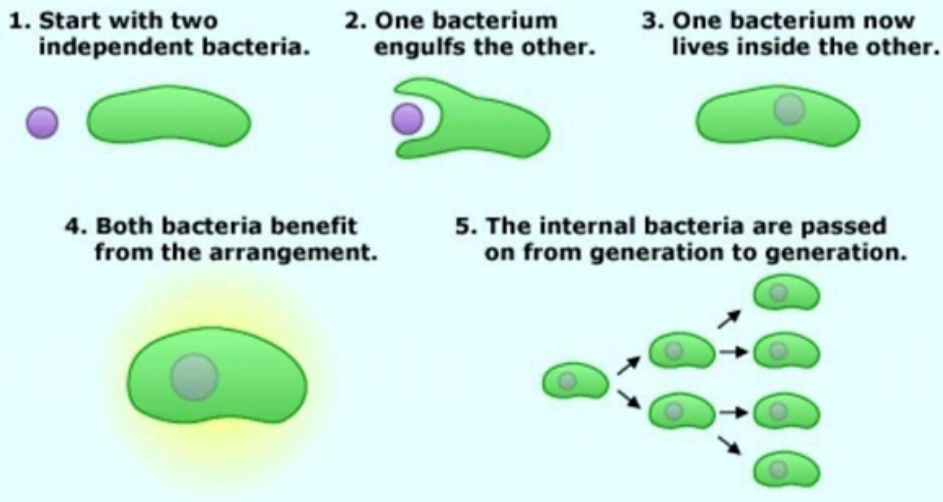
Origin of Eukaryotic Cells

Endosymbiotic Theory

- Chloroplasts and Mitochondria evolved from symbiotic bacteria that lived inside of a larger prokaryote.
- Also known as Symbiogenesis

Domain Eukaryota

Endosymbiosis in a nutshell:



<https://en.wikipedia.org/wiki/Symbiogenesis>

Domain Eukaryota

- Chloroplasts evolved from endocytosis of cyanobacteria
- Mitochondria evolved from endocytosis of aerobic bacteria.

Type of Wastewater

Components

- Pathogens
- Salts
- Acids
- Bases
- Metals
- Toxic inorganic compounds
- Organic matter
- Suspended solids

7

Type of Wastewater

• Types

1. Domestic wastewater from households
2. Municipal wastewater from communities (also called sewage)
3. Industrial wastewater from industrial activities
4. Agricultural wastewater from agricultural activities
5. Storm water

146-Evolution of 3rd atmosphere?(ppt34)

Third Atmosphere

- Constant rearrangement of continental plates
- Inflow and outflow of CO₂ from large continental carbonate stores
- O₂ synthesized in photosynthesis was being used in reducing materials
- Gradually, it was shifted to oxidizing atmosphere instead of reducing
- 541 million years ago, O₂ made stable 15% of atmosphere.

Google



structure and function of...



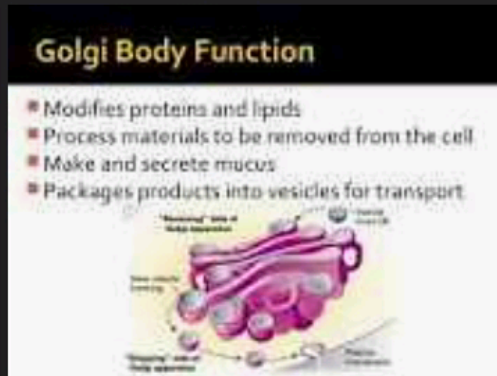
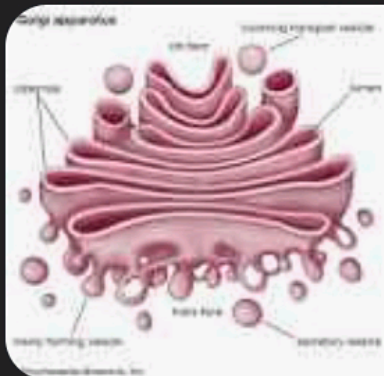
Images

Wikipedia

Pdf

Class 9

Class 11



The Golgi apparatus is a central intracellular membrane-bound organelle with key functions in trafficking, processing, and sorting of newly synthesized membrane and secretory proteins and lipids. To best perform these functions, Golgi membranes form a unique stacked structure.

<https://www.ncbi.nlm.nih.gov/pmc>

Golgi Structure and Function in Health, Stress, and Diseases - PMC - NCBI



Discover



Search



Collections

What is an invasive species explain?



An invasive species is an introduced, nonnative organism (disease, parasite, plant, or animal) that begins to spread or expand its range from the site of its original introduction and that has the potential to cause harm to the environment, the economy, or to human health.

